Solve with us 2.1 - Not Graded

This assignment will not be graded and is only for practice.

1. Key Points:

Calculating minors and cofactors of a matrix:

- If A is a square matrix, then the minor of the entry in the i-th row and j-th column (denoted by M_{ij}) is the determinant of the submatrix formed by deleting the i-th row and j-th column.
- The ij-th cofactor (denoted by C_{ij}) is defined to be $(-1)^{i+j} M_{ij}$.

Calculating the determinant:

- $det(A) = \sum_{i=1}^{n} a_{ij}(-1)^{i+j} M_{ij} = \sum_{i=1}^{n} a_{ij} C_{ij}$

Let us consider a 3×3 matrix A as follows:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Calculating M_{11} and C_{11}:

• Find out the sub-matrix by deleting the first row and the first column:

$$\begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & a_{22} & a_{23} \\ \cdot & a_{32} & a_{33} \end{bmatrix}$$

• Calculating the determinant of the sub-matrix:

$$det\left(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}\right) = a_{22}a_{33} - a_{32}a_{23}$$

- $M_{11} = a_{22}a_{33} - a_{32}a_{23}$
- $C_{11} = (-1)^{1+1}(a_{22}a_{33} - a_{32}a_{23}) = a_{22}a_{33} - a_{32}a_{23}$

Calculating M_{23} and C_{23} :

• Find out the sub-matrix by deleting the second row and the third column:

$$\begin{bmatrix} a_{11} & a_{12} & \cdot \\ \cdot & \cdot & \cdot \\ a_{31} & a_{32} & \cdot \end{bmatrix}$$

• Calculating the determinant of the sub-matrix:

$$det\left(\begin{bmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{bmatrix}\right) = a_{11}a_{32} - a_{31}a_{12}$$

- $M_{23} = a_{11}a_{32} - a_{31}a_{12}$
- $C_{23} = (-1)^{2+3}(a_{11}a_{32} - a_{31}a_{12}) = -a_{11}a_{32} + a_{31}a_{12}$

Calculating the determinant of A:

- Expanding with respect to the first row:

  $$det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$$ det(A) = a11(−1)1+1M11 + a12(−1)1+2M12 + a13(−1)1+3M13

- Expanding with respect to the second row:

  $$det(A) = a_{21}(-1)^{2+1}M_{21} + a_{22}(-1)^{2+2}M_{22} + a_{23}(-1)^{2+3}M_{23}$$ det(A) = a21(−1)2+1M21 + a22(−1)2+2M22 + a23(−1)2+3M23

- Expanding with respect to the third row:

  $$det(A) = a_{31}(-1)^{3+1}M_{31} + a_{32}(-1)^{3+2}M_{32} + a_{33}(-1)^{3+3}M_{33}$$ det(A) = a31(−1)3+1M31 + a32(−1)3+2M32 + a33(−1)3+3M33

  The determinant can also be calculated by expanding along columns.

- Expanding with respect to the first column:
  $$det(A) = a_{11}(-1)^{1+1}M_{11} + a_{21}(-1)^{2+1}M_{21} + a_{31}(-1)^{3+1}M_{31}$$ det(A) = a11(−1)1+1M11 + a21(−1)2+1M21 + a31(−1)3+1M31

Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$ A = −12−12000−11. Find out M_{23} M23.

[Hint: Find out the sub-matrix by deleting the second row and the third column: $\begin{bmatrix} -1 & 2 & \cdot \\ \cdot & \cdot & \cdot \\ -1 & 0 & \cdot \end{bmatrix}$ −1⋅−12⋅0⋯

Calculate the determinant of the sub-matrix.]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$ A = −12−12000−11. Find out C_{32} C32.

[Hint: Find out the sub-matrix by deleting the third row and the second column: $\begin{bmatrix} -1 & \cdot & 0 \\ 2 & \cdot & -1 \\ \cdot & \cdot & \cdot \end{bmatrix}$ −12⋯⋯0−1⋅

Calculate the determinant of the sub-matrix to find M_{32} M32. To find the cofactor C_{23} C23 the sign has to be taken care of as follows: $C_{32} = (-1)^{3+2}M_{32}$ C32 = (−1)3+2M32]

No, the answer is incorrect.
Score: 0

Accepted Answers:
(Type: Numeric) -1
1 point

Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$. Find out the determinant of A.

[Hint: Expanding with respect to the first row:
$$det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$$
Expanding with respect to any row will also give you the answer]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -2
1 point
1 point
Consider the following square matrices:

$$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$$

Choose the set of correct options.
Hint:
- B can be obtained by interchanging the first two rows.
- C can be obtained by multiplying 2 to the first row of B.

☐ $det(B) = det(A)$.
☐ $det(A) = -det(B)$.
☐ $det(C) \neq -2det(B)$.
☐ $det(C) = -2det(A)$

No, the answer is incorrect.
Score: 0
Feedback:
• Sign of the determinant changes due to interchange of two rows.
• If a scalar c is multiplied with a row of a matrix, then the determinant of the new matrix will be c times the determinant of the earlier matrix.
Accepted Answers:
$det(A) = -det(B)$.

$det(C) \neq -2det(B)$det(C) = −2det(B).
$det(C) = -2det(A)$det(C) = −2det(A)

2. Key Points:

Cramers' Rule: (This rule is applied to any system of linear equations with nn equations and nn variables. Here we recall the method for a system of linear equations with $3$3 equations and $3$3 variables.)

- Consider a system of linear equations as follows: $\begin{matrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{matrix}$ a11x1 + a12x2 + a13

x3 = b1a21x1 + a22x2 + a23x3 = b2a31x1 + a32x2 + a33x3 = b3

- Let the matrix representation of the above system be $Ax = b$Ax = b, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$A = $\begin{bmatrix} \text{a11a21a31a12a22a32a13a23a33} \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = $\begin{bmatrix} \text{x1x2x3} \end{bmatrix}$, and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$b = $\begin{bmatrix} \text{b1b2b3} \end{bmatrix}$.

- Let A_{x_i} Axi be the matrix obtained by replacing the ii-th column of AA $(i.e., \begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix})$(i.e., $\begin{bmatrix} \text{a1ia2ia3i} \end{bmatrix}$) by bb, for $i = 1, 2, 3$i = 1, 2, 3.

$A_{x_1} = \begin{bmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{bmatrix}$Ax1 = $\begin{bmatrix} \text{b1b2b3a12a22a32a13a23a33} \end{bmatrix}$

$A_{x_2} = \begin{bmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{bmatrix}$Ax2 = $\begin{bmatrix} \text{a11a21a31b1b2b3a13a23a33} \end{bmatrix}$

$A_{x_3} = \begin{bmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{bmatrix}$Ax3 = $\begin{bmatrix} \text{a11a21a31a12a22a32b1b2b3} \end{bmatrix}$

If $det(A) \neq 0$det(A) = 0, then the solutions to the above system are $x_i = \frac{det\ A_{x_i}}{det\ A}$xi = det Adet Axi, for $i = 1, 2, 3$i = 1, 2, 3. i.e., $x_1 = \frac{det\ A_{x_1}}{det\ A}$x1 = det Adet Ax1, $x_2 = \frac{det\ A_{x_2}}{det\ A}$x2 = det Adet Ax2, $x_3 = \frac{det\ A_{x_3}}{det\ A}$x3 = det Adet Ax3

Consider a system of linear equations $\begin{aligned} x_1 + x_3 &= 1 \\ -x_1 + x_2 - x_3 &= 1 \\ -x_2 + x_3 &= 1 \end{aligned}$

Let matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Let A_{x_i} be the matrix obtained by replacing the i-th column of A (i.e., $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$) by b, for $i = 1, 2, 3$.

Use the above information to answer the following questions(5,6):

1 point

Choose the set of correct options

- [ ] $A_{x_1} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$
- [ ] $A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$
- [ ] $A_{x_2} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$
- [ ] $A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

$A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

1 point

Choose the set of correct options about the solutions to this system.

- [ ] $x_1 = -2$.
- [ ] $x_2 = -2$.
- [ ] $x_3 = 3$.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$x_1 = -2$.

$x_3 = 3$.

3. Key Points:

Calculating the inverse of a matrix A:

- Find out the cofactor matrix C, whose ij-th element is C_{ij}: the ij-th cofactor of A.
- Adjoint of A is transpose of C.
- Calculate the determinant of A and check whether it is non-zero or not.
- If determinant is non-zero then the inverse of A exists and is given by

$$A^{-1} = \frac{1}{det(A)} adj(A).$$

1 point

Which of the following is the cofactor matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$?

[Hint: Recall the steps to calculate the cofactors, given in Key Points 1.]

- ○ $\begin{bmatrix} 1 & 1 & 1 \\ -1 & -3 & -1 \\ -2 & 0 & 2 \end{bmatrix}$
- ○ $\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$
- ○

$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$

○

$\begin{bmatrix} 1 & -1 & -1 \\ -1 & 3 & -1 \\ -2 & 0 & 2 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$

1 point

Which of the following is the adjoint matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$?

[Hint: Find the transpose to the cofactor matrix.]

○

$\begin{bmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 1 & -1 & 2 \end{bmatrix}$

○

$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$

○

$\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$

○

$\begin{bmatrix} 1 & -1 & -2 \\ -1 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$

Find out the determinant of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$.

[Hint: Recall the steps to calculate the determinant, given in Key Point 1.]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point

Which of the following is the inverse matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$?

[Hint: If determinant is non-zero then the inverse of A exists and is given by $A^{-1} = \frac{1}{det(A)} adj(A)$.]

- ◯ $\frac{1}{2}\begin{bmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 1 & -1 & 2 \end{bmatrix}$
- ◯ $\frac{1}{2}\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$
- ◯ $\frac{1}{2}\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$
- ◯ $\frac{1}{2}\begin{bmatrix} 1 & -1 & -2 \\ -1 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\frac{1}{2}\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$

1 point

Choose the set of correct options.

Hint:

- $A^{-1} = \frac{1}{det(A)} adj(A)$.
- $det(A^{-1}) = \frac{1}{det(A)}$

☐ If $A = A^{-1}$, then $det(A)$ must be 1.

☐ If $A = A^{-1}$, then A must be the identity matrix.

☐ If $A = A^{-1}$, then $A^2 = I$.

☐